

EXPERIMENT 7

SELECT with Various Clauses – GROUP BY, HAVING, ORDER BY

AIM:

To view the records from the tables using SELECT commands with Group By, Having, Order By

DESCRIPTION:

GROUP BY – HAVING:

- The GROUP BY clause groups a set of rows into a set of summary rows by values of columns or expressions. The GROUP BY clause returns one row for each group. In other words, it reduces the number of rows in the result set.
- The GROUP BY clause is used with aggregate functions such as SUM, AVG, MAX, MIN, and COUNT. The aggregate function that appears in the SELECT clause provides the information about each group.
- The GROUP BY clause is an optional clause of the SELECT statement. To filter the groups returned by GROUP BY clause, you use a HAVING clause.

ORDER BY:

When you use the SELECT statement to query data from a table, the result set is not sorted in any orders. To sort the result set, you use the ORDER BY clause. The ORDER BY clause allows you to:

- Sort a result set by a single column or multiple columns.
- Sort a result set by different columns in ascending or descending order.

QUESTIONS

GROUP BY – HAVING

1. Display the gender and number of patients for each gender, considering only genders having **more than one patient**.
2. Display the Doctor ID and number of appointments for each doctor, considering only doctors having **at least one appointment**.

ORDER BY

3. Retrieve the **Patient Name, Gender and Phone Number** of all patients in ascending order of **Patient ID**.
4. List the doctors in the order of **oldest doctor first** based on their Date of Birth.

OUTPUTS:

1)

```
238 • SELECT Gender,
239         COUNT(*) AS Number_of_Patients
240 FROM Patient
241 GROUP BY Gender
242 HAVING COUNT(*) > 1;
243
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Gender	Number_of_Patients
▶	M	2
	F	2

2)

```
245 • SELECT Doctor_ID,
246         COUNT(*) AS Number_of_Appointments
247 FROM Appointment
248 GROUP BY Doctor_ID
249 HAVING COUNT(*) >= 1;
250
```

Result Grid |  |  Filter Rows: | Export:  | Wrap Cell Content: 

	Doctor_ID	Number_of_Appointments
▶	D001	1
	D002	1
	D003	1
	D004	1

3)

```
252 • SELECT Patient_Name, Gender, Phone_No
253 FROM Patient
254 ORDER BY Patient_ID ASC;
255
```

Result Grid			
Filter Rows:			
Export:  Wrap Cell C			
	Patient_Name	Gender	Phone_No
▶	Arun Kumar	M	9876543210
	Priya	F	9876543211
	Rahul	M	9876543212
	Sneha	F	9876543213

4)

```
---
257 • SELECT *
258 FROM Doctor
259 ORDER BY DOB ASC;
260
```

Result Grid							
Filter Rows:							
Edit:    Export/Import:   Wrap Cell Cont							
	Doctor_ID	Doctor_Name	Gender	DOB	Specialization	Phone_No	Department
▶	D003	Kumar	M	1982-11-15	Orthopedics	9876500003	Orthopedics
	D001	Ravi	M	1985-04-12	Cardiology	9876500001	Emergency
	D002	Priya	F	1988-07-20	Neurology	9876500002	Neurology
	D004	Anitha	F	1990-02-25	Pediatrics	9876500004	Pediatrics
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

RESULT:

The records from the tables are displayed using SELECT commands with GROUP BY, HAVING and ORDER BY.